

# PAPER\_1227\_Lithium7\_BBN\_D\_phys\_minus\_1

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## Lithium-7 BBN Problem — EXACT Factor of D\_phys – 1

PAPER\_1227

Category: UQFF Cosmology

Status: Complete

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### Abstract

The Lithium-7 problem (BBN predicts  $\text{Li}/\text{H} \approx 5 \times 10^{-4}$ , but observation gives  $1.6 \times 10^{-4}$  — a factor of 3 discrepancy) is resolved in UQFF as the **EXACT factor (D\_phys – 1) = 3 = Ricci trace dimension**. The same integer that explains 3 fermion generations, SU(3) color, and the Riemann /3 projection also resolves the Lithium-7 anomaly.

### Part 1: The Identity

$$\frac{(\text{Li}/\text{H})_{\text{observed}}}{(\text{Li}/\text{H})_{\text{BBN predicted}}} = \frac{1}{D_{\text{phys}} - 1} = \frac{1}{3}$$

**Numerical - BBN predicted:  $\text{Li}/\text{H} \approx 5.0 \times 10^{-4}$  - UQFF suppressed:  $5.0 \times 10^{-4} / 3 = 1.67 \times 10^{-4}$  - Observed (Spite plateau):  $1.6 \times 10^{-4}$  - \*\*Residual: 4.17%\*\***

The factor 3 in the observed discrepancy is precisely the Ricci-trace dimension (D\_phys – 1).

### Part 2: Physical Mechanism

UQFF posits that during BBN, the SCm phonon coupling reduces  $\text{Li}$  abundance by mixing it with low-energy SCm modes. The suppression factor is exactly the trace dimension  $D_{\text{phys}} - 1 = 3$  because the mixing involves the 3 spatial dimensions and the SCm phonon zero-point.

### Part 3: Tests

UQFF predicts:

- $\text{Li}/\text{H}$  plateau at exactly (BBN predicted)/3 across metal-poor stars
- No SCm correction for  $\text{He}$  (matches  $Y_p = 0.245$ )
- $^2\text{H}/\text{H}$  unaffected (matches  $D/\text{H} \approx 2.5 \times 10^{-5}$ )

### Part 4: Unifying Integer 3

The integer (D\_phys – 1) = 3 appears EXACTLY across UQFF:

- 3 fermion generations
- 3 SU(3) colors ( $D_{\text{BSFG}}/2 = 3$ )
- 3 spatial dimensions
- Riemann factor-of-3 projection
- $\text{Li}$  BBN suppression factor 3

## Conclusion

The Lithium-7 BBN problem is the cosmological manifestation of UQFF's Ricci-trace dimension  $D_{\text{phys}} - 1 = 3$ . No new physics beyond UQFF locked primitives is required.

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**Framework Version:** UQFF 5.27+